The Essence of Statistics

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Thanks SM Shovan Sir

Applications of statistics

- The State : Gross wealth, population, per capita income, defence, agriculture, policy.
- Economics: What, how, for whom to produce?
- Business Management and Industry: size, # of employee, investment, maximize profit
- Social and Natural Sciences: Mortality, population growth, illiteracy rate, economic con
- Biology and Medicine: Body temperature, heart rate, blood pressure, Glucose level.
- Research: New knowledge discovery, pattern analysis, prediction.

What is statistics?

Definition

- Statistics is the discipline that concerns the collection, analysis and interpretation of data.

1. First step is the problem statement : What range of ages of student study in BSc program in RUET?

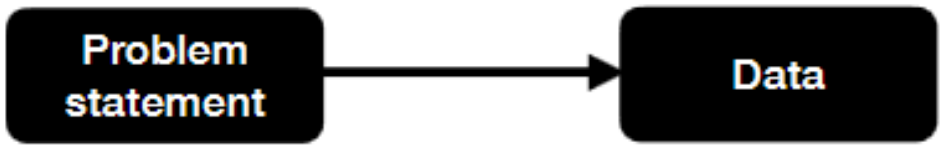
2. Data collection: 23, 20, 26, 24, …., 18, 19, 27

3. Analysis: (Sorting) 18,19, 20, … , 23, 24, 26, 27

4. Interpretation: (27-18) = 9 is the range.

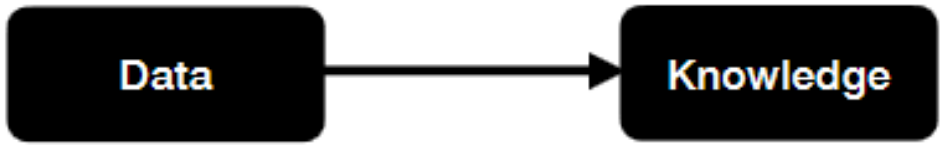
Difference between statistics and machine learning

Statistics



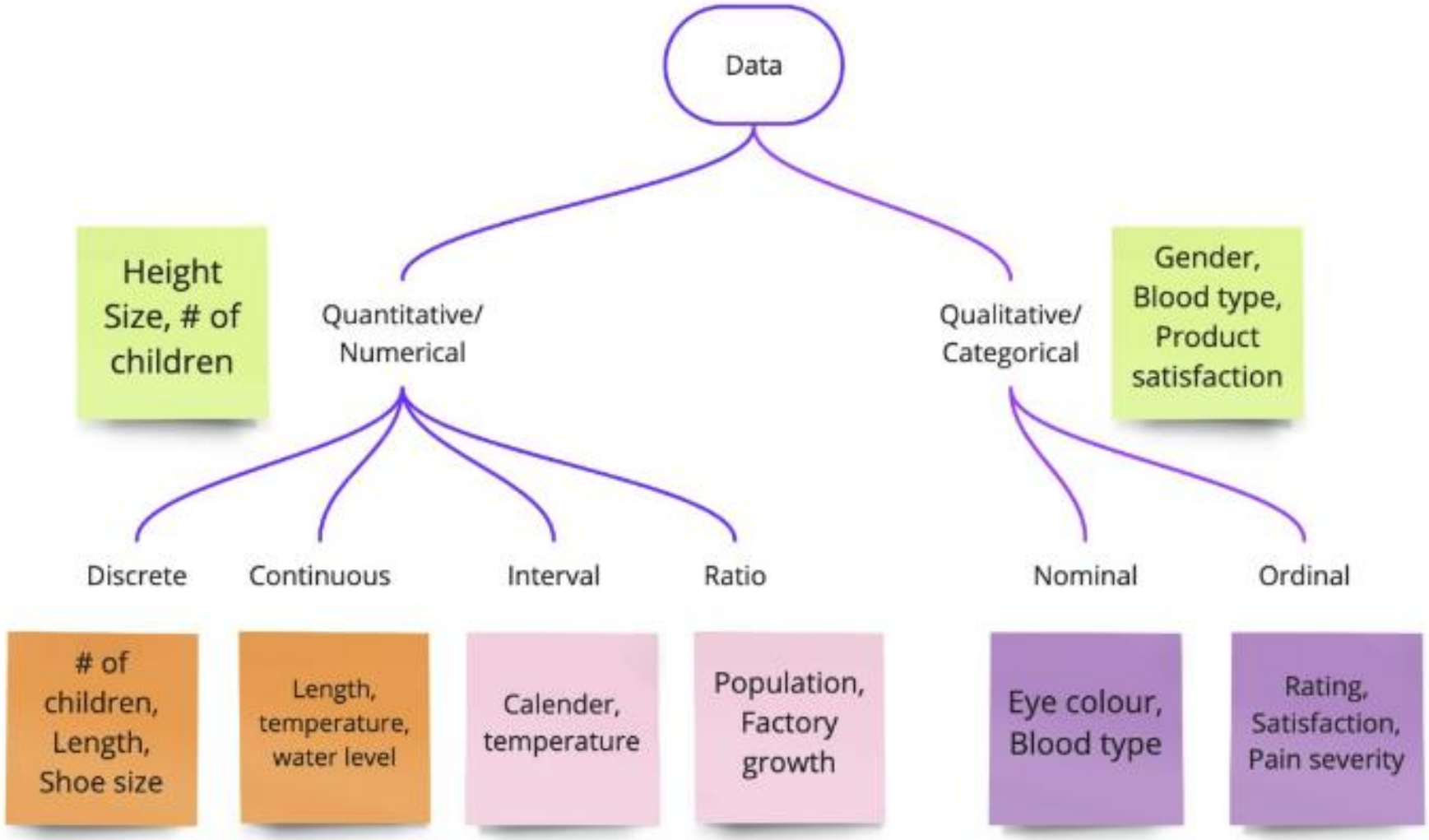
1. Illiteracy rate
2. Company losses

Machine learning



1. Super shop data
2. Weather prediction

Types of data



Quantitative / Numerical Data

- Quantitative data consists of numerical values that can be measured or counted. It answers “How much?” or “How many?”

- 📌 Types of Quantitative Data:
  - 1️⃣ Discrete Data (Countable, Whole Numbers) [Finite Range]
    - Example: Number of students in a class (e.g., 30)
    - Example: Number of books on a shelf (e.g., 12)
  - 2️⃣ Continuous Data (Measurable, Can Have Decimals) [Infinite Range]
    - Example: Height of a person (e.g., 5.8 ft)
    - Example: Temperature (e.g., 36.5°C)

Interval and Ratio

- 1 **Interval Data** (Meaningful Differences, No True Zero)
  - Example: Temperature in Celsius (0°C does not mean "no temperature")
  - Example: A clock’s 0:00 (midnight) does not mean “no time exists.”

- 2 **Ratio Data** (Meaningful Differences, True Zero Exists)
  - Example: Height, Weight, Age (0 means "no height," "no weight")
  - Example: Salary (Someone earning $0 means no income)

Qualitative (Categorical) Data

- Qualitative data represents **descriptive, non-numerical** information. It answers **"What type?"** or **"Which category?“**

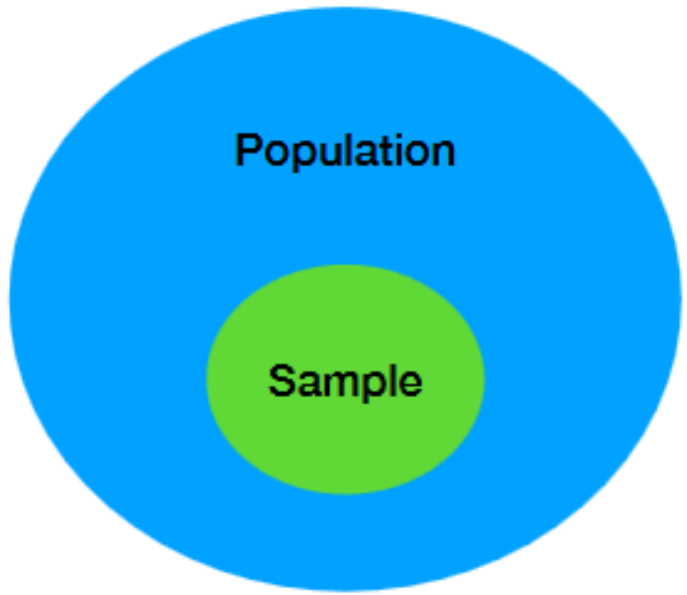
- 📌 Types of Qualitative Data:
  - **Nominal Data** (Labels, No Order)
  - Categories with **no ranking or meaningful order**
  - Example:
    - Eye Color: **Blue, Green, Brown**
    - Gender: **Male, Female, Other**
    - Blood Type: **A, B, AB, O**

  - **Ordinal Data** (Ordered Categories)
  - Categories **have a meaningful order**, but the difference between them is **not measurable**
  - Example:
    - Customer Satisfaction: **Satisfied > Neutral > Dissatisfied**
    - Education Level: **Primary < High School < Bachelor < Master**
    - Movie Ratings: **1-star, 2-star, 3-star, 4-star, 5-star**

Population & Sampling

- Population (N): refers to the collection of all elements under a specific context
- Sample (n): means a subgroup of members of population chosen for participation in a study

- It's not always possible to collect or process data for population
- Sampling must make sure that data is not biased



Population

- A population (N) is the entire group of individuals, items, or data points that we want to study.
- Example: All students in a university, Every citizen in a country, All products in a factory
- Why Not Study the Whole Population?
  - Too large to collect data from everyone
  - Expensive and time-consuming
  - Difficult to process huge amounts of data

Sample

- A sample (n) is a subset of the population chosen for study. Instead of studying everyone, we collect data from a representative group.
- Example: Surveying 500 students from a university of 50,000 students, Testing 100 products from a 1,000,000-product batch
- Why Use Sampling?
  - Saves time & cost
  - Easier to analyze
  - Helps draw conclusions about the entire population

Importance of Unbiased Sampling

- A **sample must be representative of the population** to avoid **bias** (skewed or misleading results).

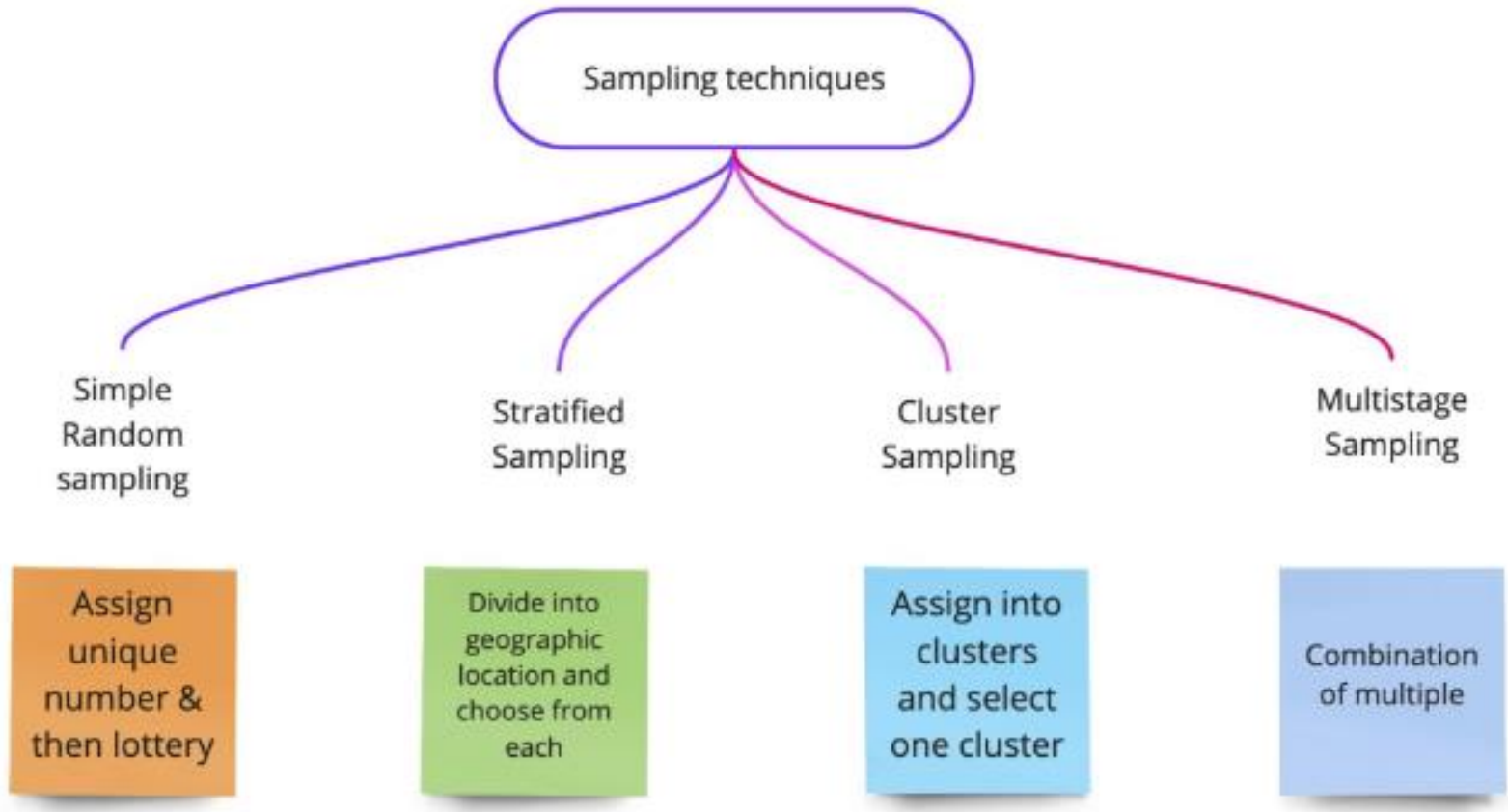
✓ **Good Sampling Example:**

Randomly selecting **students from different departments** in a university survey

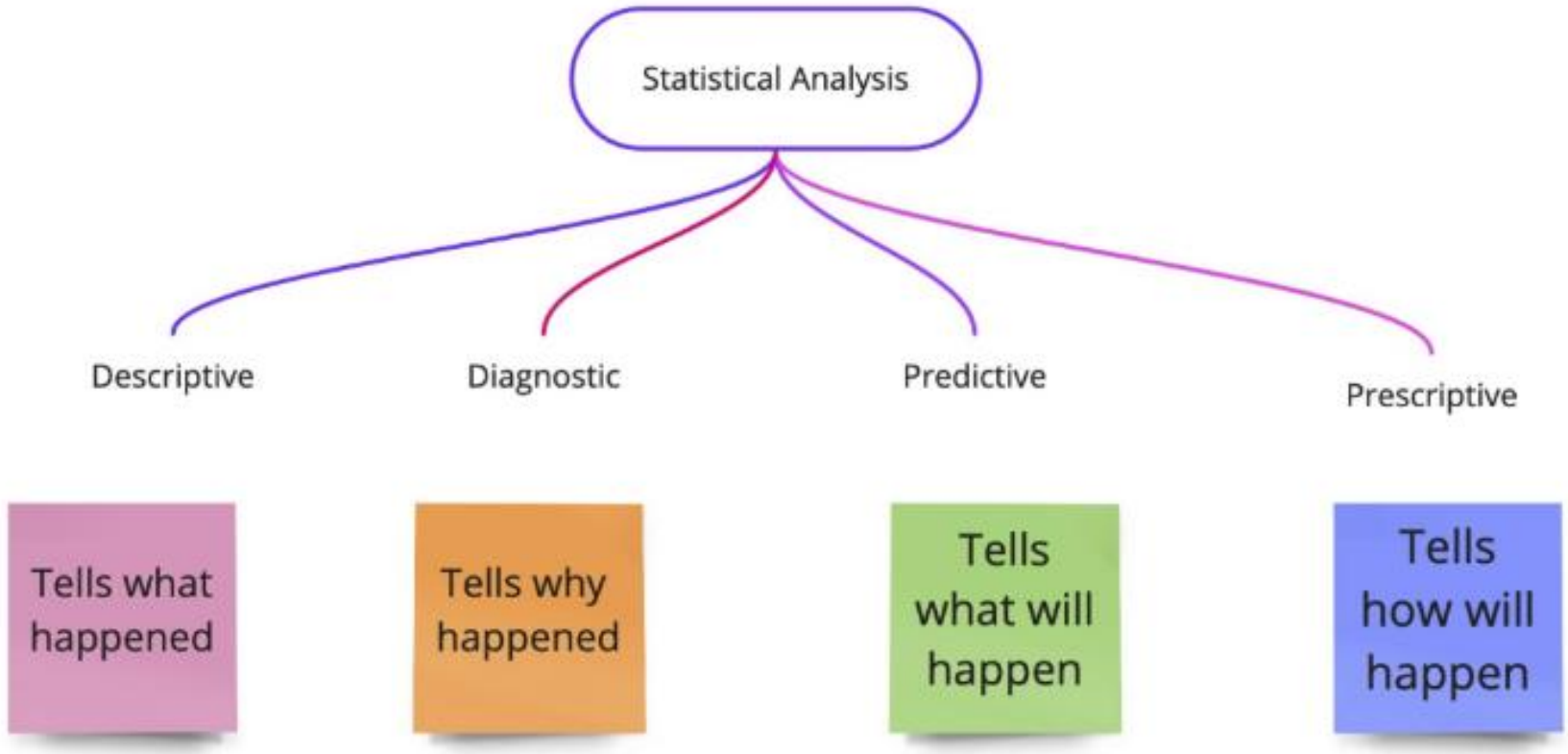
✗ **Bad Sampling Example (Biased):**

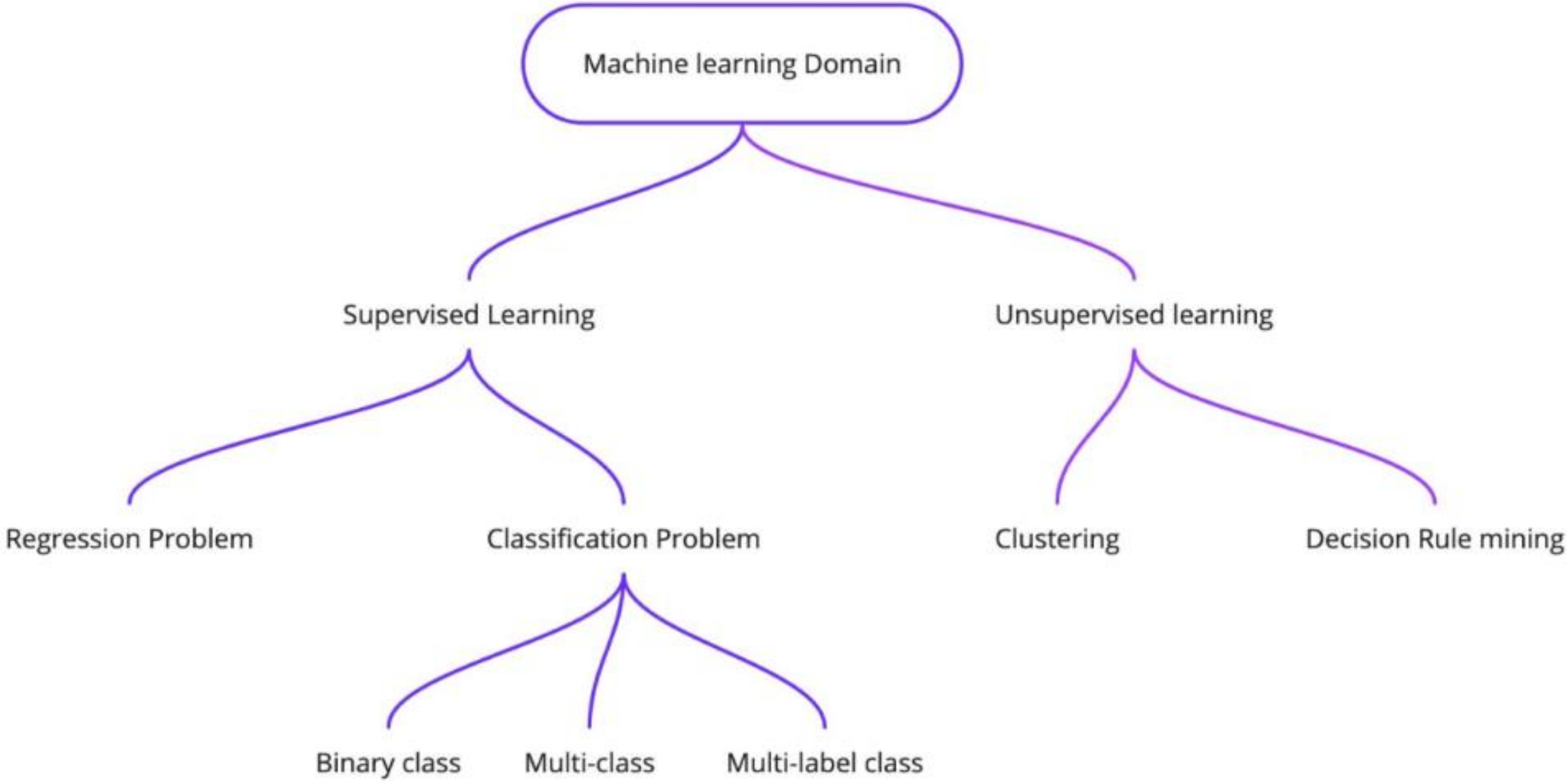
Only surveying **top students**, which does not represent all students

Sampling Techniques



Types of statistical analysis




Machine learning Domain
Supervised Learning
Unsupervised learning
Regression Problem
Classification Problem
Clustering
Decision Rule mining
Binary class
Multi-class
Multi-label class